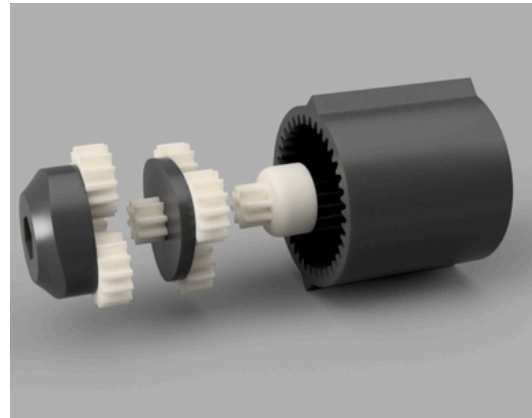
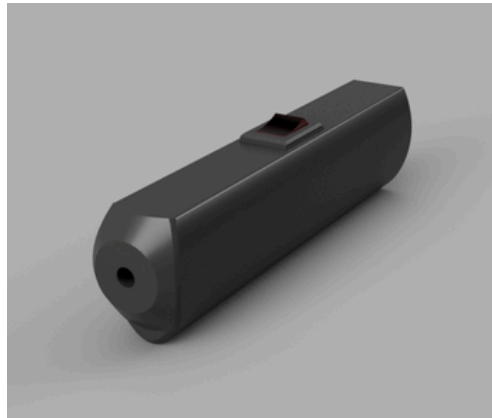
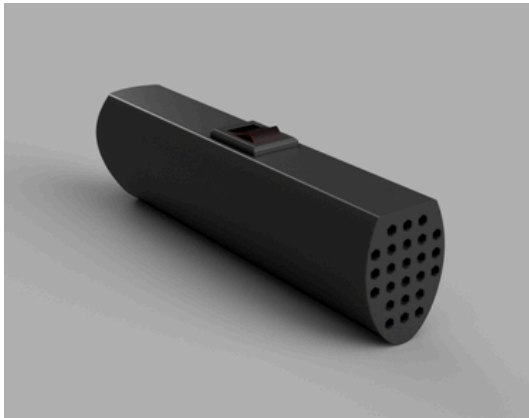


3D PRINTED HANDHELD ELECTRIC SCREWDRIVER

(INDEPENDENT)



Goals

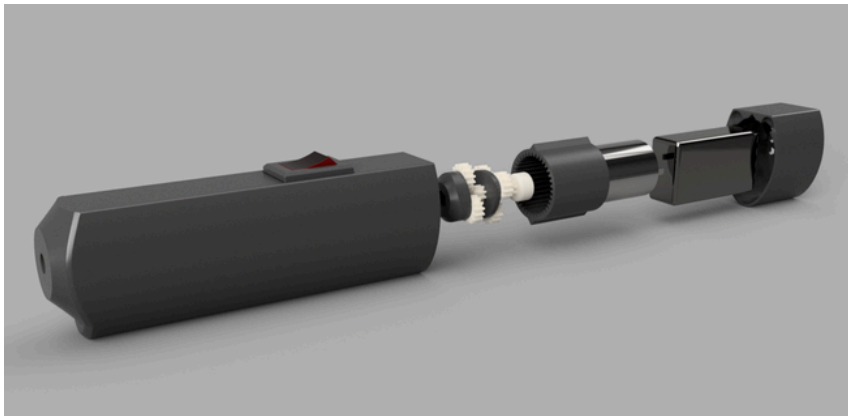
- To design and fabricate a mostly 3D printed handheld electric screwdriver
- Design for easy use with standard bits
- Allow for easy recharging

Mechanical

- Used Onshape to design every mechanical and structural component
- 3D printed every component except electronics in PLA
- Created a custom planetary gearbox for more torque

Electrical

- Used a reverse polarity rocker switch to direct a 12V DC motor powered by an enclosed battery
- Wires are soldered together inside the outer shell
- The battery easily disconnects for charging using jumper cables



Design Process Challenges

- Overcame the challenge of designing a functional 3D-printed planetary gearbox, learning gear ratios, alignment, and torque distribution
- Gained hands-on experience with soldering and electronics, learning how to wire and connect components for a functional circuit